

"Title","Authors","Year","DOI or URL","Abstract","Source"
"AI-Enabled Automation for Completeness Checking of Privacy Policies","Orlando Cejas Amaral; Sallam Abualhaija; Damiano Torre; Mehrdad Sabetzadeh; Lionel C. Briand","2022","10.1109/TSE.2021.3124332","Technological advances in information sharing have raised concerns about data protection... In this paper, we propose AI-based automation for the completeness checking of privacy policies... our approach resulted in a precision of 92.9% and recall of 89.8%, significantly improving over a keyword-based baseline ¹ ² .","IEEE Transactions on Software Engineering"

"NLP-based Automated Compliance Checking of Data Processing Agreements against GDPR","Orlando Cejas Amaral; Muhammad Ilyas Azeem; Sallam Abualhaija; Lionel C. Briand","2023","10.1109/TSE.2023.3288901","When the entity processing personal data (the processor) differs from the one collecting personal data (the controller), checking the compliance of DPAs is challenging... We propose an automated solution that leverages NLP technologies to check a given DPA against GDPR requirements... The approach achieved ~89.1% precision and 82.4% recall, outperforming an off-the-shelf NLP baseline by ~20 percentage points in accuracy ³ ⁴ .","IEEE Transactions on Software Engineering"

"ML-Based Compliance Verification of Data Processing Agreements against GDPR","Orlando Cejas Amaral; Sallam Abualhaija; Lionel C. Briand","2023","10.1109/RE57278.2023.00015","Verifying DPA compliance entirely manually is time-consuming and error-prone. This paper proposes a machine learning-based strategy for GDPR compliance checking in DPAs by predicting the presence of key information types. On 30 real DPAs, the approach detected 483/582 actual violations (83.0% recall) with 83.9% precision, comparable to a rule-based method. The study discusses factors (data availability, expert input) affecting when ML vs. rule-based solutions are preferable ⁵ ⁶ .","Proc. 31st IEEE Requirements Engineering Conference (RE 2023)"

"Optimizing Automated Compliance Checking with Ontology-Enhanced NLP: Case in the Fire Safety Domain","Yian Chen; Huixian Jiang","2024","10.1016/j.jenvman.2024.123320","Fire safety compliance checking (FSCC) requires interpreting numerous spatial rules in fire codes. This study proposes an automated framework combining ontology and NLP to interpret fire regulation rules. Using pre-trained language models, concepts and relationships are extracted, a domain-specific ontology is established, and spatial knowledge is transformed into logical clauses based on semantic consistency, improving the rule interpretation and compliance checking accuracy ⁷ ⁸ .","Journal of Environmental Management"

"Automated Code Compliance Checking Research Based on BIM and Knowledge Graph","Junlong Peng; Xiangjun Liu","2023","10.1038/s41598-023-34342-1","Traditional drawing reviews are manual and inefficient. This study uses BIM and knowledge graph technology to automatically check building code compliance. Regulatory provisions are transformed into a machine-readable structured format using NLP to build a human-machine-

readable knowledge graph. BIM model data are exported and mapped to rule representations; compliance is then automatically checked. An example verifies the method's feasibility, addressing "manual dependency" and "inefficiency" in code compliance reviews ⁹ ¹⁰ .", "Scientific Reports (Nature)"

"An Automated Fire Code Compliance Checking Jointly Using BIM and Natural Language Processing", "Yukang Wang; Yue Liu; Haozhe Cai; Jia Wang; Xiaoping Zhou", "2023", "10.3390/fire6090358", "Fire code compliance checking ensures building safety by verifying designs against fire regulations. This paper presents a BIM-based ACC (Automated Compliance Checking) system that uses NLP to analyze spatial geometric relationships of building components. The framework divides ACC into building model parsing, code knowledge translation (using NLP to formalize spatial stipulations), and compliance result reporting. In tests on various building types, the proposed system demonstrated high recall and efficient, accurate compliance checking ¹¹ ¹² .", "Fire (MDPI)"

"Automated BIM Compliance Check through a Large Language Model Combined with Deep Learning and Ontology", "Nanjiang Chen; Xuhui Lin; Hai Jiang; Yi An", "2024", "10.3390/buildings14071983", "Ensuring AEC projects comply with complex regulations is challenging. Traditional manual checking is inefficient. NLP and deep learning have improved rule interpretation, but need extensive data and tuning. This study introduces an ACC framework integrating a Large Language Model (LLM), deep learning, and ontology. LLM few-shot learning reduces the need for large labeled datasets. Deep learning first classifies regulatory text to aid the LLM's structured information extraction. This combined approach significantly improves automated processing of regulatory texts, offering a scalable solution for compliance checking in construction ¹³ ¹⁴ .", "Buildings (MDPI)"

"Invariant Signature, Logic Reasoning, and Semantic NLP-Based Automated Building Code Compliance Checking (I-SNACC) Framework", "Jin Wu; Xiaorui Xue; Jiansong Zhang", "2023", "10.36680/j.itcon.2023.001", "Manual building code compliance checks are costly and error-prone. With BIM, automation is feasible but existing methods have limited adaptability across different codes. This paper proposes I-SNACC, a framework requiring minimal user input for fully automated compliance checking using invariant signatures, logic reasoning, and semantic NLP. A prototype was tested on Chapter 10 of the 2015 International Building Code using two real projects, achieving 95.2% precision and 100% recall in detecting non-compliance. The framework increases automation and reduces manual effort compared to state-of-the-art methods ¹⁵ ¹⁶ .", "Journal of Information Technology in Construction (ITcon)"

"Automating Compliance Evidence Extraction with Machine Learning", "Sunday O. Olatunji; Abdulrahman S. Almuhaideb; Abdulaziz S. Alhuwaishil; Ahmad S. Alarfaj; Khalifah W. Alzawaimel; Hassan A. Alsaleem; Abdulaziz A. Alsudais; Naeem M. Owaida; Fahd A. Alhaidari; Sghaier R.

Chabani", "2025", "10.1007/978-981-97-8588-9_3", "This study reviews the intersection of cybersecurity and compliance, highlighting automation's benefits in compliance evidence processing. It explores how machine learning can analyze documents to verify if they meet required evidence standards. Compliance assessments typically demand manual review of voluminous documentation. By automating with ML, significant time and cost savings and accuracy improvements are achieved. The paper provides a thorough summary of

recent ML-driven compliance automation approaches, discussing algorithms, datasets, and outcomes in the domain ¹⁷ ¹⁸ .", "Proc. 1st Int. Conf. on Creativity, Technology, and Sustainability (CCTS 2024)"

"A Platform-Based NLP-Driven Strategy for Digitalising Regulatory Compliance Processes (Built Environment)", "Ruben Kruiper; Bimal Kumar; Richard Watson; Farhad Sadeghineko; Alasdair J. G. Gray; Ioannis Konostas", "2024", "10.1016/j.aei.2024.102653", "Digitalising regulatory compliance has gained momentum, especially after the 2017 Grenfell fire. This paper presents i-ReC (intelligent Regulatory Compliance), a comprehensive platform that applies NLP, machine learning, and Semantic Web technologies to process actual regulatory documents for compliance checking. The approach handles challenges like ambiguous language, inconsistent terminology, and multi-word expressions. A semantic search engine was developed and tested on 420 UK building regulation documents, consistently outperforming other engines on retrieval tasks. The work underscores the importance of extracting requirements from large regulatory corpora to enable end-to-end automated compliance checking of building designs ¹⁹ ²⁰ .", "Advanced Engineering Informatics"

¹ ² [2106.05688] AI-enabled Automation for Completeness Checking of Privacy Policies

<https://arxiv.org/abs/2106.05688>

³ ⁴ arxiv.org

<https://arxiv.org/pdf/2209.09722>

⁵ ⁶ orbilu.uni.lu

[https://orbilu.uni.lu/bitstream/10993/55408/1/ML-](https://orbilu.uni.lu/bitstream/10993/55408/1/ML-based%20Compliance%20Verification%20of%20Data%20Processing%20Agreements%20against%20GDPR.pdf)

[based%20Compliance%20Verification%20of%20Data%20Processing%20Agreements%20against%20GDPR.pdf](https://orbilu.uni.lu/bitstream/10993/55408/1/ML-based%20Compliance%20Verification%20of%20Data%20Processing%20Agreements%20against%20GDPR.pdf)

⁷ ⁸ Optimizing automated compliance checking with ontology-enhanced natural language processing: Case in the fire safety domain - PubMed

<https://pubmed.ncbi.nlm.nih.gov/39547025/>

⁹ ¹⁰ Automated code compliance checking research based on BIM and knowledge graph | Scientific Reports

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¹¹ ¹² An Automated Fire Code Compliance Checking Jointly Using Building Information Models and Natural Language Processing

<https://www.mdpi.com/2571-6255/6/9/358>

¹³ ¹⁴ Automated Building Information Modeling Compliance Check through a Large Language Model Combined with Deep Learning and Ontology

<https://www.mdpi.com/2075-5309/14/7/1983>

¹⁵ ¹⁶ ITcon paper: Invariant Signature, Logic Reasoning, and Semantic Natural Language Processing (NLP)-Based Automated Building Code Compliance Checking (I-SNACC) Framework [2023-1]

<https://www.itcon.org/paper/2023/1>

¹⁷ ¹⁸ Automating Compliance Evidence Extraction with Machine Learning | SpringerLink

https://link.springer.com/chapter/10.1007/978-981-97-8588-9_3

